MATHEMATICAL FOUNDATIONS OF STOCHASTIC DIFFUSION: ANALYTICAL SOLUTIONS FOR COMPLEX GEOMETRIES (WEDGE & RING)

Abstract: This technical report presents the analytical derivation of the Fokker-Planck equation for particle diffusion in non-Euclidean geometries. By leveraging Bessel functions and Fourier series, we establish the theoretical benchmarks used to validate the numerical stochastic simulations in the accompanying repository.

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- Diffusion in a wedge: For a Brownian particle diffusing in an infinite wedge with absorbing boundaries (see figure), calculate the probability that a particle starting from (r_0, θ_0) (where θ_0 is the angle with the horizontal) is absorbed by the horizontal boundary of the wedge.

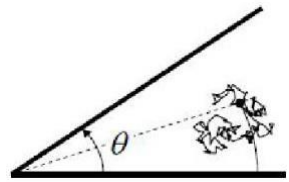


Figure. Diffusion in a wedge. An infinite wedge forms an angle θ with the horizontal. The boundaries are absorbing. What is the probability that a Brownian particle starting at (r_0, θ_0) is absorbed by the horizontal boundary?

Solution:

Since this is a Brownian particle moving in two dimensions, we know that the probability distribution for the particle's position at time t satisfies the diffusion equation or the 2D Fokker-Planck equation:

$$\frac{\partial P}{\partial t} = D\nabla^2 P,$$

Which, for our appropriate reference system, would be:

$$\frac{\partial P(r,\theta,t)}{\partial t} = D\left[\frac{\partial^2 P}{\partial r^2} + \frac{1}{r}\frac{\partial P}{\partial r} + \frac{1}{r^2}\frac{\partial^2 P}{\partial \theta^2}\right],$$

Now, we solve by proposing separation of variables:

$$P(r,\theta,t) = R(r)\Theta(\theta)T(t),$$

Substituting and rearranging to separate variables:

$$\frac{\partial R(r)\Theta(\theta)T(t)}{\partial t} = D\left[\frac{\partial^2 R(r)\Theta(\theta)T(t)}{\partial r^2} + \frac{1}{r}\frac{\partial R(r)\Theta(\theta)T(t)}{\partial r} + \frac{1}{r^2}\frac{\partial^2 R(r)\Theta(\theta)T(t)}{\partial \theta^2}\right],$$

$$R(r)\Theta(\theta)\frac{dT(t)}{dt} = D\left[\Theta(\theta)T(t)\frac{d^2R(r)}{dr^2} + \Theta(\theta)T(t)\frac{1}{r}\frac{dR(r)}{dr} + R(r)T(t)\frac{1}{r^2}\frac{d^2\Theta(\theta)}{d\theta^2}\right],$$

$$\frac{R(r)\Theta(\theta)}{D}\frac{dT(t)}{dt} = \Theta(\theta)T(t)\frac{d^2R(r)}{dr^2} + \Theta(\theta)T(t)\frac{1}{r}\frac{dR(r)}{dr} + R(r)T(t)\frac{1}{r^2}\frac{d^2\Theta(\theta)}{d\theta^2},$$

$$\frac{1}{DT(t)}\frac{dT(t)}{dt} = \frac{1}{R(r)}\left[\frac{d^2R(r)}{dr^2} + \frac{1}{r}\frac{dR(r)}{dr}\right] + \frac{1}{r^2\Theta(\theta)}\frac{d^2\Theta(\theta)}{d\theta^2},$$

Now, by linear independence, we have:

$$\frac{1}{DT(t)}\frac{dT(t)}{dt} = -\lambda,$$

$$\frac{1}{R(r)}\left[\frac{d^2R(r)}{dr^2} + \frac{1}{r}\frac{dR(r)}{dr}\right] + \frac{1}{r^2\Theta(\theta)}\frac{d^2\Theta(\theta)}{d\theta^2} = -\lambda,$$

$$\frac{1}{\Theta(\theta)}\frac{d^2\Theta(\theta)}{d\theta^2} = -\left[r^2\lambda + \frac{r^2}{R(r)}\left[\frac{d^2R(r)}{dr^2} + \frac{1}{r}\frac{dR(r)}{dr}\right]\right],$$

Therefore, the separated differential equations are:

$$\frac{1}{DT(t)}\frac{dT(t)}{dt} = -\lambda,$$

$$\frac{1}{\Theta(\theta)}\frac{d^2\Theta(\theta)}{d\theta^2} = -\gamma,$$

$$r^2\lambda + \frac{r^2}{R(r)}\left[\frac{d^2R(r)}{dr^2} + \frac{1}{r}\frac{dR(r)}{dr}\right] = \gamma,$$

We rearrange to show that the second equation is of harmonic oscillator type and the third of Bessel type, simplifying the rest of the solution:

$$\frac{d\ln T(t)}{dt} = -\lambda D,$$

$$\frac{d^2\Theta(\theta)}{d\theta^2} + \gamma\Theta(\theta) = 0,$$

$$r^2\frac{d^2R(r)}{dr^2} + r\frac{dR(r)}{dr} + (r^2\lambda - \gamma)R(r) = 0$$

Given that the solutions are well-known (Bessel and harmonic oscillator type), we will directly use the general solutions and evaluate the boundary conditions:

$$\ln T(t) = -\lambda t + c \Longrightarrow T(t) = Ce^{-\lambda Dt},$$

$$\Theta(\theta) = Asen\left(\sqrt{\gamma}\theta\right) + Bcos\left(\sqrt{\gamma}\theta\right),$$

$$R(r) = DJ_{\sqrt{\gamma}}\left(\sqrt{\lambda}r\right) + EY_{\sqrt{\gamma}}\left(\sqrt{\lambda}r\right),$$

Now, we evaluate the boundary conditions:

$$\Theta(0) = Asen\left(\sqrt{\gamma}(0)\right) + Bcos\left(\sqrt{\gamma}(0)\right) = 0 \Longrightarrow B = 0,$$

$$\Theta(\theta_0) = Asen\left(\sqrt{\gamma}\theta_0\right) + Bcos\left(\sqrt{\gamma}\theta_0\right) = 0 \Longrightarrow \sqrt{\gamma} = \frac{n\pi}{\theta_0} | n\epsilon\mathbb{Z},$$

$$R(r_0) = DJ_{\sqrt{\gamma}}\left(\sqrt{\lambda}r_0\right) + EY_{\sqrt{\gamma}}\left(\sqrt{\lambda}r_0\right) = 0 \Longrightarrow E = 0,$$

Since $r \rightarrow \infty, R(r) \rightarrow 0$ we have the quantization $\sqrt{\lambda} = \frac{j_{\sqrt{\gamma},k}}{r_0}$, where j are the zeros of the Bessel function; therefore, the solution is given by:

$$P(r,\theta,t) = \sum_{n,k=0}^{\infty} F_{nm} sen\left(\frac{n\pi}{\theta_0}\theta\right) J_{\frac{n\pi}{\theta_0}}\left(\frac{j_{\frac{n\pi}{\theta_0},k}}{r_0} r\right) e^{-\left(j_{\frac{n\pi}{\theta_0},k}\right)^2 Dt},$$

Now, the flux is given by:

$$J(r,t) = -\frac{D}{r}\frac{\partial P(\theta,t,r)}{\partial\theta}\bigg|_{\theta=0},$$

$$J_{total} = \int_0^{\infty} J(r,t)dr,$$

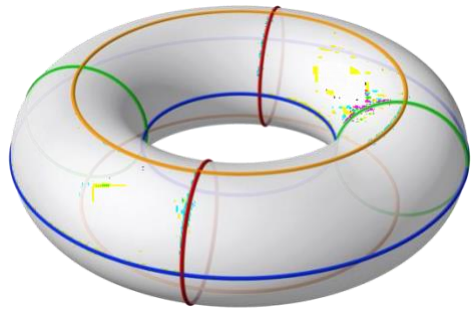
$$G(t) = \int_0^t |J_{total}(t')|dt',$$

$$\tau = \int_0^t t'|J_{total}(t')|dt',$$

Finally, the probability that the particle has not been absorbed up to time t (survival function), $S(t)$, is calculated as:

$$S(t) = 1 - \int_0^t |J_{total}(t')|dt',$$

- Consider a large, circular, tubular ring of circumference $2l$ containing a chemical of concentration $c(x,t)$ dissolved in water. Let x be the arc length parameter with $0 < x < 2l$.



If the concentration of the chemical is initially given by $f(x)$, then $c(x,t)$ satisfies the initial-boundary value problem:

$$c_t = Dc_{xx} \qquad 0 < x < 2l, t > 0,$$

$$c(0,t) = c(2l,t), \qquad c_x(0,t) = c_x(2l,t),$$

$$c(x,0) = f(x),$$

Find the concentration $c(x,t)$.

We propose $c(x,t) = X(x)T(t)$:

$$\frac{\partial c(x,t)}{\partial t} = D\frac{\partial^2 c(x,t)}{\partial x^2} \Rightarrow X(x)\frac{dT(t)}{dt} = DT(t)\frac{d^2X(x)}{dx^2},$$

$$\Rightarrow \frac{1}{DT(t)}\frac{dT(t)}{dt} = \frac{1}{X(x)}\frac{d^2X(x)}{dx^2},$$

Now we have separated variables, and the only way for the differential equation to hold is for them to be equal to a constant:

$$\frac{1}{DT(t)}\frac{dT(t)}{dt} = -\lambda,$$

$$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} = -\lambda,$$

We solve each one and apply the boundary conditions:

$$\frac{d\ln T(t)}{dt} = -\lambda D,$$

$$\frac{d^2X(x)}{dx^2} + \lambda X(x) = 0,$$

It can be seen that the second equation is of the simple harmonic oscillator type, whose solution is well-known. One could propose exponentials and find the roots of the polynomial, and then express the exponentials as sines and cosines, or we can simply propose the solution and evaluate it:

$$\ln T(t) = -\lambda t + c \Longrightarrow T(t) = Ce^{-\lambda Dt},$$

$$X(x) = Asen\left(\sqrt{\lambda}x\right) + Bcos\left(\sqrt{\lambda}x\right),$$

We evaluate the boundary conditions:

$$X(0) = Asen\left(\sqrt{\lambda}(0)\right) + Bcos\left(\sqrt{\lambda}(0)\right) = B = X(2l) = Asen\left(\sqrt{\lambda}2l\right) + Bcos\left(\sqrt{\lambda}2l\right),$$

$$B = \frac{Asen\left(\sqrt{\lambda}2l\right)}{1 - cos\left(\sqrt{\lambda}2l\right)},$$

$$X'(0) = \sqrt{\lambda}\left[Acos\left(\sqrt{\lambda}(0)\right) - Bsen\left(\sqrt{\lambda}(0)\right)\right] = \sqrt{\lambda}A = X'(2l),$$

$$X'(2l) = \sqrt{\lambda}\left[Acos\left(\sqrt{\lambda}(2l)\right) - Bsen\left(\sqrt{\lambda}(2l)\right)\right] = \sqrt{\lambda}A,$$

$$A = -\frac{Bsen\left(\sqrt{\lambda}(2l)\right)}{1 - cos\left(\sqrt{\lambda}(2l)\right)} = -\frac{Asen^2\left(\sqrt{\lambda}2l\right)}{\left[1 - cos\left(\sqrt{\lambda}2l\right)\right]^2},$$

$$A\left[1 - cos\left(\sqrt{\lambda}L\right)\right]^2 + Asen^2\left(\sqrt{\lambda}L\right) = 0$$

$$A2cos\left(\sqrt{\lambda}L\right) = 0 \Longrightarrow \sqrt{\lambda} = \frac{\left(n + \frac{1}{2}\right)\pi}{2l} | n\epsilon\mathbb{Z}, \Longrightarrow B = \frac{Asen\left(\sqrt{\lambda}2l\right)}{1 - cos\left(\sqrt{\lambda}2l\right)} = (-1)^n A$$

$$T(t) = Ce^{-\left(\frac{\left(n+\frac{1}{2}\right)\pi}{2l}\right)^2 Dt},$$

$$X_n(x) = A_n\left[sen\left(\frac{\left(n + \frac{1}{2}\right)\pi}{2l}x\right) + (-1)^n cos\left(\frac{\left(n + \frac{1}{2}\right)\pi}{2l}x\right)\right],$$

Therefore :

$$c(x,t) = \sum_{n=0}^{\infty} G_n\left[sen\left(\frac{\left(n + \frac{1}{2}\right)\pi}{2l}x\right) + (-1)^n cos\left(\frac{\left(n + \frac{1}{2}\right)\pi}{2l}x\right)\right]e^{-\left(\frac{\left(n+\frac{1}{2}\right)\pi}{L}\right)^2 Dt},$$

We evaluate the last boundary condition:

$$c(x,0) = \sum_{n=0}^{\infty} G_n \left[sen\left(\frac{\left(n+\frac{1}{2}\right)\pi}{2l} x \right) + (-1)^n cos\left(\frac{\left(n+\frac{1}{2}\right)\pi}{2l} x \right) \right] = f(x),$$

$$G_n = \frac{1}{l} \int_0^{2l} f(x) \left[sen\left(\frac{\left(n+\frac{1}{2}\right)\pi}{2l} x \right) + (-1)^n cos\left(\frac{\left(n+\frac{1}{2}\right)\pi}{2l} x \right) \right] dx$$

Thus obtaining G_n for the expression:

$$c(x,t) = \sum_{n=0}^{\infty} G_n \left[sen\left(\frac{\left(n+\frac{1}{2}\right)\pi}{2l} x \right) + (-1)^n cos\left(\frac{\left(n+\frac{1}{2}\right)\pi}{2l} x \right) \right] e^{-\left(\frac{\left(n+\frac{1}{2}\right)\pi}{L} \right)^2 Dt},$$